Open problems:
Physical vector spaces for quantum mechanics

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

From the brief

1. **Completeness of the Lie algebra.** *Research.* Definition 1 admits pathological cases: a Hilbert space with the defining observables of a qubit ("too many vectors") and a Schwartz space with position as the only defining observable ("too few observables"). "Addressing this gap is an open problem"; the brief expects a "final definition" once it is solved. Partial answer (`further-results.tex`, Sec. 2): the second case is not a quantum vector space, since with position alone $\mathcal{S}(\mathbb{R})$ is not complete (`FurtherResults.not_isQuantumVectorSpace_posOnly`). The first case is one (`FurtherResults.isQuantumVectorSpace_pair`, for $\mathbb{C}^2 \otimes \mathbb{C}^2$), and irreducibility excludes it: every continuous operator that commutes with all defining observables is a multiple of the identity (`FurtherResults.not_isIrreducible_pair`). This is a form of Roberts's condition (J. Math. Phys. 7 (1966) 1097, p. 1099), and Examples 3 and 4 satisfy it (`FurtherResults.isIrreducible_schwartz`, `FurtherResults.isIrreducible_qubit`). Open: whether irreducibility is the condition the brief is looking for, and how it relates to taking the smooth vectors of an irreducible unitary group representation as V. (Brief, Sec. 5, "Completeness of the Lie algebra"; Sec. 6.)

2. **Lie algebras determine physical systems.** *Research.* "Two systems are physically equivalent if and only if their Lie algebras of observables are equivalent" is asserted, and is what makes QSTO:V consistent with [B]STDIST. (Brief, Theorem 5, proof; Log, "Differences", last item.)

3. **Ensembles on a quantum vector space.** *Research.* QSTO:V uses density operators ρ and $S(\rho) = -\operatorname{tr}(\rho \log \rho)$, and Sec. 5 calls the set of ensembles "the most natural space to consider", but no ensemble space over V is defined. In particular, whether the ensembles induce on the pure states the quotient topology of V, under which the pure states are second countable exactly when V is, is open. *Partial answer* (`further-results.tex`, Sec. 4). For Example 3 the positive trace-one Schwartz operators of Keyl, Kiukas and Werner are a natural ensemble space: they are the density operators with all moments $\operatorname{tr} \rho N^j$ finite (`FurtherResults.IsSchwartzDensity`), they contain every pure state of V (`FurtherResults.isSchwartzDensity_proj`), and the von Neumann entropy is finite (`FurtherResults.IsSchwartzDensity.vonNeumannEntropy_ne_top`) and continuous on them, by Wehrl's theorem (`QuantumMechanics.tendsto_vonNeumannEntropy`,

`FurtherResults.tendsto_vonNeumannEntropy_of_tendsto_numberMoment`). The question about the quotient topology remains open. (Brief, Sec. 3, QSTO:V; Sec. 5.)

4. **Field theories.** *Research.* "For a field theory, we would expect the defining observables to be at least countable"; no field-theory example of a quantum vector space is given. (Brief, Sec. 5, "Number of defining observables".)

5. **Which observables generate unitary groups.** *Research.* Since symmetric and self-adjoint coincide on V, "not all self-adjoint operators generate a unitary transformation", and "to understand which ones the integrability of the system must be checked". A characterization, analogous to complete vector fields in Hamiltonian mechanics, is not given. (Brief, Sec. 5, "Symmetric vs self-adjoint".)

6. **Which changes of frame.** *Research.* The proof of Theorem 5 gets [B]WDQ from the premise that "the defining observables of A must be observables in B and vice-versa", which holds exactly when the two frames give the same topology on V (`topology_eq_iff_continuous`). The brief does not say which changes of frame satisfy it. For Example 3 the change of frame $y = \tan(\frac{\pi}{2} \operatorname{erf} x)$, which the brief "Unphysicality of Hilbert spaces for quantum mechanics" uses against [B]WDQ for QSTO:H, does not: it takes a Gaussian out of $\mathcal{S}(\mathbb{R})$ (`FurtherResults.not_forall_frameUnitary_mem_range`), and the position of the second frame, carried back to the first, is not defined at that Gaussian (`FurtherResults.toLp_gaussSchwartz_not_mem_domain`; see `further-results.tex`, Sec. 1). So QSTO:V gives [B]WDQ only for a class of changes of frame that map V onto itself, as Carcassi, Calderón and Aidala already require for Schwartz spaces (arXiv:2308.06669v3, Appendix, Sec. VIII). The class is not specified. The restriction cannot simply be dropped: a wave function on $\mathbb{R}$ has finite position variance in every smooth coordinate only if it vanishes outside a bounded set; then, unless it is zero, its Fourier transform does not, and the free evolution destroys the cut-off at once. This is proved in `briefs/HilbertUnphysicality/further-results.tex`, Sec. 2 (`HilbertUnphysicality.FurtherResults.not_forall_frames_freeEvolution`), and with the brief "Unphysicality of Hilbert spaces for quantum mechanics" (`HilbertUnphysicality.quantum_no_cutoff`). Open: which group of changes of frame should be part of the data of a quantum vector space, and how it relates to the Lie algebra of the defining observables. For plain changes of variable $f \mapsto f \circ g$ on $\mathcal{S}(\mathbb{R})$, those that map $\mathcal{S}(\mathbb{R})$ into itself are characterized by Galbis and Jordá (Rev. Mat. Iberoam. 34 (2018), Thm. 2.3); the unitary changes of frame carry the extra factor $\sqrt{g'}$. *Partial answer* (`further-results.tex`, Sec. 3). With a group F of changes of frame acting on V by linear bijections that preserve the inner product, the premise holds for every pair of frames in F if and only if every element of F acts continuously on V (`FurtherResults.isFrameCompatible_iff_continuous`). Then every change of frame in F is a unitary equivalence (`FurtherResults.IsFrameCompatible.unitaryEquiv`). It suffices that the transformed defining observables lie in the algebra of the original ones (`FurtherResults.isFrameCompatible_of_mem_algebra`); the translations of Example 3 are an instance (`FurtherResults.isFrameCompatible_translate`). The choice of F remains open. (Brief, Theorem 5, proof; Log, "Problems found in the brief", last item.)

From the formalization

None. Definition 1 and Examples 3 (in n dimensions) and 4 are checked in full.